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Chemistry Paper 3A WASSCE (SC), 2023

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Question 1A

A is $0.100 \text{ mol dm}^{-3}$ of HCl.

B is $0.030 \text{ mol dm}^{-3}$ of a trioxocarbonate (IV) salt.

(a) Put A into the burette and titrate it with 20.0 cm^3 or 25.0 cm^3 portion of B using methyl orange as indicator.

Repeat the titration to obtain concordant titre values.

Tabulate your results and calculate the average volume of A used.

Put A into the burette and titrate it with 20.0 cm^3 or 25.0 cm^3 portion of B using methyl orange as Indicator.

Repeat the titration to obtain concordant titre values.

Tabulate your results and calculate the average volume of A used.

(b) From your results and the information provided, calculate the mole ratio of the acid to the trioxocarbonate (IV) in the reaction.

(c) Given that B contains 5.0 g dm^{-3} of the hydrated trioxocarbonate (IV) salt, calculate the:

(i) concentration of anhydrous salt in B in g dm^{-3} ,

(ii) percentage of water hydration in B;

(iii) number of moles of hydrogen ions in the average titre value.

[Molar mass of an anhydrous salt in B = 106 g mol^{-1}]

Observation

This question was based on acid-base titration, and was attempted by the majority of the candidates.

In part (a), majority of the candidates correctly calculated the average volume of A used.

In part (b), majority of the candidates were unable to calculate the mole ratio of the acid to the trioxocarbonate (IV) in the reaction.

In part (c), few candidates were able to calculate the concentration of anhydrous salt in B in g dm^{-3} . Also, percentage of water of hydration in B was correctly

calculated by few candidates.

The expected answers include:

(a) Two concordant titres

Averaging

(b) Amount of acid in A = $0.1 \times \frac{V_A}{1000}$

1000

= say a mol

Amount of CO_3^{2-} in B = $0.030 \times \frac{V_B}{1000}$

1000

= Say b mol

Mole ratio = a : b

Simpler whole number ratio = A : B

(c) (i) Conc. of anhydrous salt in g dm^{-3}

= molar conc. $\times$ molar mass

= 0.030×106

= 3.18 g dm^{-3}

(correct to 3 sig. fig to score, wrong unit no score)

(ii) % of water of hydration

$\frac{\text{mass of water of hydration in dm}^3}{\text{mass of hydrated salt in dm}^3} \times 100$

Mass of water of hydration = $5.0 \text{ g dm}^{-3} - 3.18 \text{ g dm}^{-3}$

= 1.82 g

% of water of hydration = $\frac{1.82}{5.0} \times 100$

5.0

= 36.4%

(iii) 1 mole of HCl contain 1 mole of H^+

1000 cm^3 of A contain 0.1 moles H^+

$V_A \text{ cm}^3$ will contain $\frac{V_A \times 0.1}{1000}$

1000

No. of moles of H^+ = $0.0001 V_A$

Say Q mol

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